

Course Name:	Digital Design Laboratory	Semester:	III
Date of Performance:	16 / 10/ 2023	Batch No:	A2
Faculty Name:	Dr Kiran Ajetrao	Roll No:	16010122034
Faculty Sign & Date:		Grade/Marks:	___/25

Experiment No: 8
Title: 1-bit adder on VHDL

Aim and Objective of the Experiment:

To implement 1-bit adder on VHDL

COs to be achieved:

CO4: Implement digital networks using VHDL

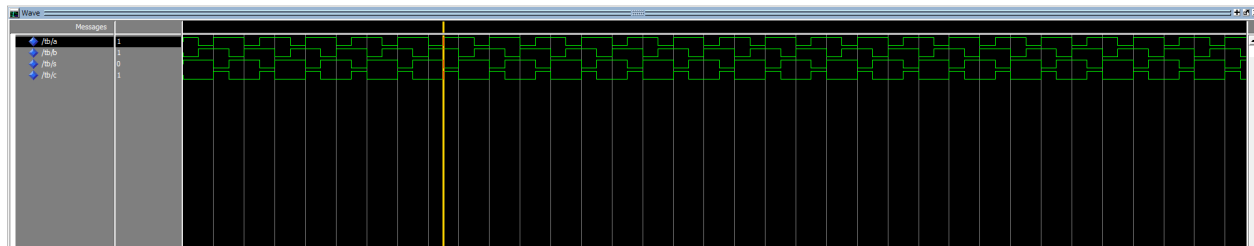
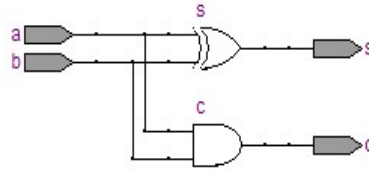
Tools used:

Quartus, ModelSim

Theory:

A 1-bit adder, a fundamental component of digital circuits, performs binary addition of two 1-bit numbers. It utilizes logic gates to generate the sum and carry-out outputs. A half-adder adds two bits without considering the carry from the previous stage, while a full-adder accounts for the carry input. Using VHDL, a hardware description language, the 1-bit adder can be designed as a combinational circuit. VHDL facilitates the creation of a structural and behavioral description of the adder. In practice, this simple unit serves as a building block for constructing larger multi-bit adders, enabling arithmetic operations in microprocessors and digital systems.

Implementation Details



Tips & Tricks

HA.vhd

```

1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity HA is
5  port(a,b:in std_logic;
6       s,c: out std_logic);
7  end HA;
8
9  architecture HA_arch of HA is
10 begin
11     s<= a xor b;
12     c<= a and b;
13 end HA_arch;

```

Tips & Tricks

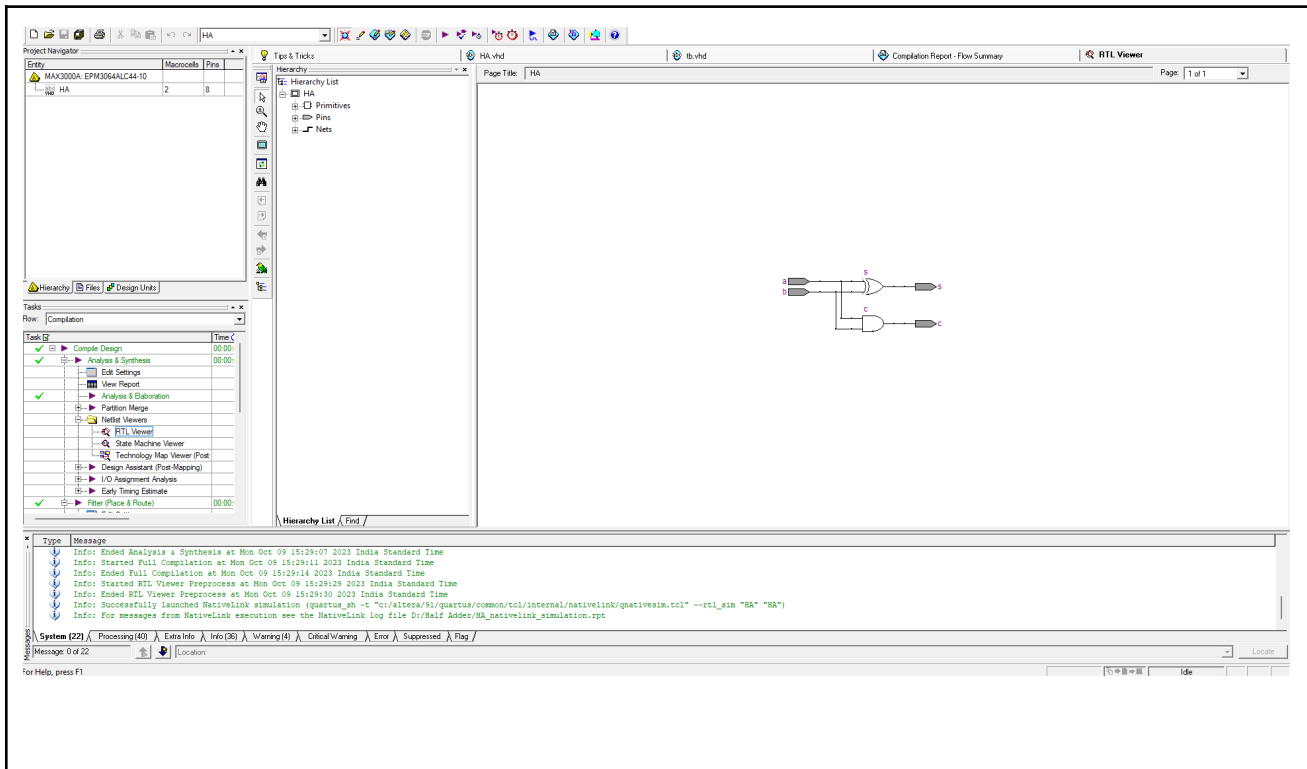
HA.vhd

tb.vhd

```

1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity tb is
5  end tb;
6
7  architecture tb_arch of tb is
8  component HA is
9  port(a,b:in std_logic;
10      s,c: out std_logic);
11 end component;
12 signal a,b,s,c:std_logic;
13 begin
14     tbm: HA port map(a,b,s,c);
15 process
16 begin
17     a<='1';
18     b<='0';
19     wait for 5ns;
20
21     a<='0';
22     b<='1';
23     wait for 5ns;
24
25     a<='1';
26     b<='1';
27     wait for 5 ns;
28 end process;
29 end tb_arch;
30
31

```



Post Lab Subjective/Objective type Questions:

1. How can 1-bit adder be used to implement a 4-bit adder?

A 1-bit adder, also known as a half-adder, can be used as a building block to create a 4-bit adder. To build a 4-bit adder, you would cascade four 1-bit adders together, with each 1-bit adder handling one bit of the input and output.

1. Define the Inputs and Outputs: A 4-bit adder takes two 4-bit inputs, A and B, and produces a 4-bit sum (S) and a carry-out (Cout).

2. Cascading Half-Adders: To add two 4-bit numbers, you start by adding the least significant bits (LSBs) first. So, you connect the LSBs of A and B to the inputs of the first half-adder, resulting in a 1-bit sum (S0) and a carry-out (Cout0). The sum S0 is the LSB of the output, and Cout0 is then carried to the next stage.

3. Repeat the Process: Connect the next bit (bit 1) of A and B to the inputs of the second half-adder. Additionally, connect Cout0 from the first half-adder to one of the inputs of the second half-adder. This simulates adding bit 1 along with the carry-out from the previous stage. You get another 1-bit sum (S1) and carry-out (Cout1).

4. Continue Cascading: Continue this process for bits 2 and 3, connecting them to the inputs of the third and fourth half-adders and taking carry-outs from the previous stages. This will give you S2, S3, Cout2, and Cout3.

5. **Final Output:** The final 4-bit sum (S) is composed of S3, S2, S1, and S0 from the individual half-adders, while the carry-out Cout3 represents the carry-out of the entire 4-bit addition.

2. What is VHDL used for?

VHDL can be used for designing hardware and for creating test entities to verify the behavior of that hardware. VHDL is used as a design entry format by a variety of EDA tools, including synthesis tools such as Quartus Prime Integrated Synthesis, simulation tools, and formal verification tools.

Conclusion:

Successfully implemented 1 bit adder on VHDL.

Signature of faculty in-charge with Date: